

A. AdaBoost

Let $\text{corr}(\mathbf{x}, \mathbf{x}')$ denote the inner product (or unnormalized correlation) of two vectors $\mathbf{x}$ and $\mathbf{x}'$.

- (1) Prove that the distribution vector $(D_{t+1}(1), \dots, D_{t+1}(m))$ defined by AdaBoost and the vector of components $y_i h_t(x_i)$ are uncorrelated.
- (2) Fix $\epsilon \in (0, 1/2)$. Let the training sample be defined by m points in the plane with $\frac{m}{4}$ negative points all at coordinate $(1, 1)$, another set of $\frac{m}{4}$ negative points all at coordinate $(-1, -1)$, $\frac{m(1-\epsilon)}{4}$ positive points all at coordinate $(1, -1)$, and $\frac{m(1+\epsilon)}{4}$ positive points all at coordinate $(-1, +1)$. Describe the behavior of AdaBoost when run on this sample using boosting stumps, in particular, give the solution the algorithm returns after T rounds.

B. Boosting algorithms

- (1) Show that for all $u \in \mathbb{R}$ and integer $p > 1$, $1_{u \leq 0} \leq \Psi_p(-u)$ where $\Psi_p(u) = \max((1+u)^p, 0)$. Show that Ψ_p is convex and differentiable.
- (2) Use Ψ_p to derive a boosting-type algorithm using coordinate descent. You should give a full description of your algorithm, including the pseudocode, details for the choice of the step and direction, as well as a generalization bound.

C. Regularized Maxent

This problem studies L_2 -regularized Maxent. We will use the notation introduced in class and will denote by J_S the dual objective function minimized given a sample S :

$$J_S(\mathbf{w}) = \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \mathbb{E}_{x \sim S} [-\log p_{\mathbf{w}}[x]],$$

where $\lambda > 0$ is a regularization parameter. We will assume that the feature vector is bounded: $\|\Phi(x)\|_2 \leq r$ for all $x \in \mathcal{X}$, for some $r > 0$.

- (1) Use McDiarmid's inequality to prove that for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds:

$$\left\| \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)] - \mathbb{E}_{x \sim S} [\Phi(x)] \right\|_2 \leq \sqrt{\frac{2r^2}{m}} \left(1 + \log \sqrt{\frac{1}{\delta}} \right).$$

- (2) Let $\hat{\mathbf{w}}$ be the L_2 -regularized maxent solution for a sample S and $\mathbf{w}_{\mathcal{D}}$ the solution for an infinite sample:

$$\hat{\mathbf{w}} = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} J_S(\mathbf{w}) \quad \text{and} \quad \mathbf{w}_{\mathcal{D}} = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} J_{\mathcal{D}}(\mathbf{w}).$$

where $J_{\mathcal{D}}(\mathbf{w}) = \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \mathbb{E}_{x \sim \mathcal{D}} [-\log p_{\mathbf{w}}[x]]$. Use the definition of $\hat{\mathbf{w}}$ and $\mathbf{w}_{\mathcal{D}}$ as minimizers (use gradients) to prove that the following inequality holds:

$$\|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2 \leq \frac{\|\mathbb{E}_{x \sim S}[\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}}[\Phi(x)]\|_2}{\lambda}.$$

- (3) For any $\mathbf{w}$ and any distribution $\mathcal{Q}$ define $\mathcal{L}_{\mathcal{Q}}(\mathbf{w})$ by $\mathcal{L}_{\mathcal{Q}}(\mathbf{w}) = \mathbb{E}_{x \sim \mathcal{Q}} [-\log p_{\mathbf{w}}[x]]$. Show that

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) - \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \leq (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot \left[\mathbb{E}_{x \sim S} [\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)] \right] + \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 - \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2.$$

- (4) Use that to show that the following inequality holds for any $\mathbf{w}$:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \frac{1}{\lambda} \left\| \mathbb{E}_{x \sim S} [\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)] \right\|_2^2 + \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2.$$

- (5) Conclude by proving that for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \inf_{\mathbf{w} \in \mathbb{R}^N} \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \frac{2r^2}{\lambda m} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)^2.$$

D. Randomized Halving

In class, we showed that, in the realizable scenario (at least one expert is always correct), the number of mistakes made by Halving is upper bounded by $\log_2 N$. Here, we consider for the same realizable scenario a randomized version of Halving defined as follows.

As for Halving, let H_t denote the set of remaining experts at the beginning of round t , with $H_1 = H$ the full set of N experts. At each round, let r_t be the fraction of experts in H_t predicting 1. Then, the prediction $\hat{y}_t$ made by the algorithm is 1 with probability

$$p_t = \left\lceil \frac{1}{2} \log_2 \frac{1}{1 - r_t} \right\rceil 1_{r_t \leq \frac{3}{4}} + 1_{r_t > \frac{3}{4}},$$

0 with probability $1 - p_t$. The true label y_t is then received and H_{t+1} is derived from H_t by removing all experts who made a mistake.

- (1) Write the pseudocode of the algorithm.

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- (2) Define the potential function $\Psi_t = \log_2 |H_t|$. Let $\mu_t = 1_{y_t \neq \hat{y}_t}$, prove that for all $t \geq 1$, $\mathbb{E}[\mu_t] \leq \frac{\Psi_t - \Psi_{t+1}}{2}$.
- (3) Show that the expected number of mistakes made by randomized Halving is at most $\frac{1}{2} \log_2 N$.
- (4) (Bonus question) Prove that no randomized algorithm makes fewer than $\lfloor \frac{1}{2} \log_2 N \rfloor$ mistakes, in expectation.